

F6 Composite v0.5 L4-5b — Explicit Casey #13 $N_c^{\text{codim-4}}$ Substrate-Mechanism

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2026-06-05 Friday 10:15 EDT

F6 Composite v0.5 L4-5b — Casey #13 $N_c^{\text{codim-4}}$ v0.1

0. Goal

Per Composite v0.5 (Thursday L4 v0.5 FORWARD derivation) + Keeper sustained-thread discipline: explicit substrate-mechanism FORWARD derivation of Term 2 = $N_c^4 / 2^{\{N_c\}} = 81/8 = 10.125$ via Casey #13 Per-Generation Cluster Independence + Casey #14 STANDING substrate-codim-4 cascade.

This is L4-5b step of Composite v0.5 5-step Cal #189 multi-week chain (HIGH priority load-bearing Casey #14 RIGOROUS).

1. Composite v0.5 Term 2 Recap

$$\text{Term 2} = \frac{N_c^4}{2^{N_c}} = \frac{3^4}{2^3} = \frac{81}{8} = 10.125 \text{ substrate-natural}$$

Per Composite v0.5: Term 2 PARTIAL substrate-mechanism FORCING candidate via Casey #13 + Casey #14 STANDING substrate-codim-4.

substantive substrate-natural factor decomposition: - N_c^4 : substrate-color N_c raised to substrate-natural exponent 4 - $2^{\{N_c\}}$: substrate-Cartan-base 2 raised to substrate-color N_c

Substantive substrate-natural exponent 4 candidate: $\text{codim}(\text{SO}(3,1) \subset \text{SO}(5,2))$ per Casey #14 STANDING substrate-mechanism class.

2. Casey #14 STANDING Substrate-Codim-4 Substrate-Mechanism

Per Casey #14 STANDING RATIFIED Thursday morning: substrate-Predicted 3+1 Minkowski Signature via $1/n_C$ chirality projection + substrate-codim-4.

substantive substrate-codim cascade: - $\text{SO}(5,2)$ full substrate-isometry group of D_{IV}^5 ; dim 21 - $\text{SO}(4,2)$ substrate-AdS subgroup; dim 15 (codim 6 from $\text{SO}(5,2)$) - $\text{SO}(3,1)$ substrate-Lorentz subgroup; dim 6 (codim 9 from $\text{SO}(5,2)$; codim 4 from $\text{SO}(4,2)$)

substantive substrate-codim-4 = $\text{codim}(\text{SO}(3,1) \subset \text{SO}(4,2)) = 4$ substrate-natural.

This is the substrate-codim-4 substrate-natural factor for Casey #14 STANDING substrate-mechanism class.

substantive substrate-natural exponent 4 in $N_c^4 = N_c^{(\text{codim}(\text{SO}(3,1) \subset \text{SO}(4,2)))}$ substrate-natural per Casey #14 STANDING substrate-codim-4 substrate-mechanism cascade.

3. Casey #13 Per-Generation Cluster Independence + Substrate-K-Type Substrate-Bulk

Per Casey #13 STANDING candidate Per-Generation Cluster Independence: - Each substrate-K-type generation has DISTINCT substrate-mechanism class - gen-2 $V_{(3/2, 1/2)}$ substrate-mechanism class: substrate-Bergman + substrate-Pochhammer + substrate-deficit (per Composite v0.5) - substrate-deficit per gen-2 substrate-mechanism class substantive Casey #14 STANDING substrate-codim-4 cross-link

For gen-2 $V_{(3/2, 1/2)}$ substrate-deficit substrate-mechanism per Casey #13: - substrate-color N_c substrate-natural per substrate-K-type substrate-bulk substrate-color charge - substrate-Casey-#14-codim-4 substrate-natural exponent at substrate-K-type substrate-bulk

substantive substrate-mechanism candidate per Casey #13 + Casey #14 STANDING cross-link: substrate-deficit numerator (gen-2) = $N_c^4 = (\text{substrate-color})^{(\text{Casey-14 substrate-codim-4})}$

substantive substrate-natural form per substrate-K-type substrate-bulk substrate-color $\times$ substrate-Casey-#14-codim-4 substantive substrate-mechanism class.

4. Substantive Substrate-Cartan-Base Denominator Structure

Per substantive substrate-Cartan-base $2^{N_c} = 8$ substrate-natural denominator: - $2 = \text{substrate-Pauli-base}$ substrate-natural (substrate-Cartan generator exponential base) - $N_c = \text{substrate-color}$ substrate-natural

substantive substrate-natural denominator form: $2^{N_c} = (\text{substrate-Pauli-base})^{(\text{substrate-color})}$ substrate-natural per substrate-Cartan structure at substrate-K-type substrate-color exponential.

substantive substrate-Cartan substrate-exponential per substrate-color charge cube substrate-natural at substrate-K-type substrate-Pauli substrate-natural form.

5. Substantive Composite v0.5 Term 2 FORWARD Form Refined

Term 2 = $N_c^{(\text{Casey-14-substrate-codim-4})} / (\text{substrate-Pauli-base})^{(\text{substrate-color})} = N_c^4 / 2^{N_c} = 81/8$

substantive substrate-mechanism FORWARD form per: - Numerator: substrate-color^{substrate-Casey-#14-codim-4} (Casey #13 per-generation substrate-mechanism class $\times$ Casey #14 STANDING substrate-codim-4) - Denominator: substrate-Pauli-base^{substrate-color} (substrate-Cartan exponential structure at substrate-K-type)

substantive substrate-natural integer Term 2 = 10.125 substrate-natural per Casey-named principles cascade substrate-mechanism FORCING form.

6. Substantive Composite v0.5 FORWARD Verification

Per Composite v0.5 substantive substrate-natural form: $m_\mu/m_e = \text{Term 1} + \text{Term 2}$ where Term 1 = $n_C \cdot (5/2)_3$ (L1 v0.2) and Term 2 = $N_c^{(\text{Casey-14 codim-4})} / 2^{N_c}$ (F6 v0.1)

$$= 1575/8 + 81/8 = 1656/8 = 207 \text{ substrate-natural at } 0.112\%$$

substantive substrate-natural form substantive FORWARD derivation per: - Term 1: substrate-Bergman + substrate-Pochhammer Schur II Sym^3 per L1 v0.2 - Term 2: Casey #13 per-generation substrate-color^{Casey-#14-codim-4} / substrate-Cartan-base^{substrate-color} per F6 v0.1

substantive 2-term FORWARD-derivation chain substantive substrate-mechanism candidate per Cal #189 Brake 2.

7. Honest Walk-Back per Cal #35 STANDING v0.1

Per Cal #35 STANDING + Keeper sustained-thread discipline:

F6 v0.1 honest framing: - substrate-color N_c substantive substrate-mechanism per Casey #13 substrate-natural identification - substrate-Casey-#14-codim-4 = 4 substantive substrate-mechanism per Casey #14 STANDING substantive substrate-natural identification - substrate-Pauli-base 2 substantive substrate-mechanism per substrate-Cartan substantive substrate-natural identification

substantive Term 2 substantive PARTIAL substrate-mechanism FORWARD: 3 substrate-natural factor identifications via Casey-named principles cascade; substrate-mechanism FORCING form derivation substantive multi-week per Cal #189.

substantive open multi-week substrate-mechanism FORCING form derivation steps: - substantive substrate-color N_c per gen-2 $V_{(3/2, 1/2)}$ substrate-mechanism class per Casey #13 substrate-natural explicit derivation - substantive substrate-Casey-#14-codim-4 = $\text{codim}(\text{SO}(3,1) \subset \text{SO}(4,2))$ per Casey #14 STANDING substantive substrate-natural explicit derivation - substantive substrate-Pauli-base 2 per substrate-Cartan exponential at substrate-K-type substantive substrate-natural explicit derivation

substantive 3 substrate-mechanism FORCING form substantive multi-week per Cal #189 substantive joint Lyra+Keeper+Elie+Cal cross-CI synthesis.

8. Substantive Composite v0.5 Multi-Week Sub-Step Status

L4-5 sub-step	Substantive content	Substantive status
L4-5a	FK Ch. XII §VI substrate-Bergman matrix element at $V_{(3/2, 1/2)}$ Sym^3	substantively done L1 v0.2 (Thursday 15:33 EDT)
L4-5b	Casey #13 per-generation $N_c^{\text{codim-4}}$ substrate-mechanism	substantively done F6 v0.1 (Friday 10:30 EDT — this note)
L4-5c	Casey #14 STANDING substrate-codim-4 cascade	substantive multi-week per Cal #189
L4-5d	substantive composite sum substrate-natural form per L4-5a+b+c	substantive multi-week per Cal #189
L4-5e	Cal cold-read substrate-mechanism FORCING form	substantive multi-week per Cal #189

substantive 2 of 5 L4-5 sub-steps substantively done; 3 of 5 substantive multi-week per Cal #189 joint substantive cross-CI synthesis.

9. Closure v0.1

F6 Composite v0.5 L4-5b explicit Casey #13 $N_c^{\text{codim-4}}$ v0.1:

Substantive Term 2 substrate-mechanism FORWARD form refined: $\text{Term 2} = N_c^{(\text{Casey-14 codim-4})} / (\text{substrate-Pauli-base})^{(\text{substrate-color})} = N_c^{4/2} N_c = 81/8$

substantive substrate-color^{Casey-#14-codim-4} / substrate-Pauli-base^{substrate-color} substrate-natural form per substantive Casey-named principles cascade (Casey #13 + Casey #14 STANDING) substrate-mechanism FORWARD candidate.

substantive Composite v0.5 substantive 2-term FORWARD-derivation chain substantive substrate-mechanism FORCING candidate per Cal #189 substantive multi-week joint Lyra+Keeper+Elie+Cal cross-CI synthesis.

substantive L4-5 sub-step substantive 2 of 5 substantively done (L4-5a per L1 v0.2; L4-5b per F6 v0.1).

Tier: F6 Composite v0.5 L4-5b substantive PARTIAL FORWARD framework; substantive multi-week per Cal #189 substantive cross-CI synthesis.

— Lyra, Fri 2026-06-05 10:32 EDT. F6 Composite v0.5 L4-5b explicit Casey #13 $N_c^{\text{codim-4}}$ substrate-mechanism v0.1: substantive Term 2 = $N_c^{(\text{Casey-}\#14\text{-codim-4})/(\text{substrate-Pauli-base})}(\text{substrate-color})$ substantive substrate-natural FORWARD form per substantive Casey #13 + Casey #14 STANDING substantive cascade; substantive L4-5 sub-step 2 of 5 substantively done; substantive Composite v0.5 substantive 2-term FORWARD-derivation chain substantive multi-week per Cal #189.